

NF4反量化

实验平台

英伟达：GH200

沐曦：C500

摩尔线程：S5000

基准测试

配置bitsandbytes环境进行基准测试，对于下列形状的权重矩阵生成，调用bitsandbytes进行量化及反量化，记录反量化的结果文件及运行的时间。与后续的实现进行平均误差和最大误差的对比。

bitsandbytes测试在英伟达GH200进行。后续对比时各平台核函数先预热5次，再执行100次求平均时间。

(256, 256)

(512, 512)

(1024, 1024)

(2048, 2048)

(4096, 4096)

(8192, 8192)

(16384, 16384)

(3421, 3146)

(6578, 1236)

(7000, 7000)

python环境配置

```
conda create -n nf4 python=3.10 -y
pip install torch --index-url https://download.pytorch.org/whl/cu124
pip install bitsandbytes numpy pandas
conda install -c nvidia cuda-nvcc cuda-runtime cuda-cudart -y
cd 03_nf4_dequant/
python generate_and_benchmark_bnb.py #生成bitsandbytes基准进行对比
```

代码文件结构

因为权重文件存储太大，不上传到github中

Learning-CUDA

├── 03_nf4_dequant/

| ├── bnb_results/ # bitsandbytes 解量化输出结果

| ├── cuda_results/ # CUDA kernel 解量化输出结果

| ├── musa_results/ # MUSA 平台解量化结果

| ├── mx_results/ # MX 平台解量化结果

| ├── mexc/ # 摩尔线程平台实现代码

| ├── mx/ # 沐曦平台 kernel 实现

| ├── nv/ # CUDA kernel 实现

| ├── weight_data/ # NF4 量化后的权重测试数据

| └── generate_and_benchmark_bnb.py # 生成 NF4 测试数据并运行 bitsandbytes benchmark

- | — bnb_benchmark_results.csv # bitsandbytes benchmark 结果
- | — 总结报告.md # NF4 解量化实验总结
- | — README.md

nf4反量化理论

```
NF4[qi•]×(code2[absmax_q[block_idx]•]×absmax2•+offset)
```

英伟达平台程序执行及性能分析

```
cd nv
./run_all.sh //对于程序进行编译和对于所有权重文件运行，生成结果文件及时间
python compare_results.py //进行性能和误差分析
```

英伟达优化思路

nf4_dequant_v1:

基础nf4反量化，实现两级缩放的解量化，建立NF4查找表(NF4_LUT)，在计算中使用float进行计算,一个线程计算一个元素，NF4_LUT放入常量内存。每个线程一次处理两个 4-bit 索引，计算两个 BF16 值后打包成一个 32-bit uint32_t 一次性写入全局内存。

Shape	Block	BnB (ms)	cuda (ms)	Speedup	MAE	Max Diff
4096x4096	128	0.0905	0.0666	1.36	0.00002557	0.00390625
4096x4096	64	0.0891	0.0678	1.31	0.00001973	0.00390625
3421x3146	64	0.0925	0.0436	2.12	0.00001746	0.00390625
7000x7000	128	0.0939	0.1926	0.49	0.00001764	0.00390625
2048x2048	64	0.0910	0.0183	4.97	0.00003022	0.00390625
8192x8192	128	0.0912	0.3057	0.30	0.00002319	0.00390625
6578x1236	64	0.0895	0.0819	1.09	0.00003105	0.00390625
2048x2048	128	0.0925	0.0179	5.17	0.00001603	0.00390625
7000x7000	64	0.0909	0.2325	0.39	0.00002593	0.00390625
256x256	128	0.1064	0.0028	38.00	0.00002176	0.00195312
256x256	64	0.1144	0.0027	42.37	0.00003350	0.00390625
6578x1236	128	0.0876	0.0326	2.69	0.00001675	0.00390625
512x512	64	0.1010	0.0032	31.56	0.00001991	0.00195312
16384x16384	128	0.2973	1.6876	0.18	0.00002354	0.00390625
16384x16384	64	0.3074	1.7599	0.17	0.00002462	0.00390625

Shape	Block	BnB (ms)	cuda (ms)	Speedup	MAE	Max Diff
8192x8192	64	0.0918	0.3188	0.29	0.00001830	0.00390625
1024x1024	64	0.0960	0.0091	10.55	0.00001878	0.00390625
3421x3146	128	0.0904	0.0421	2.15	0.00001740	0.00390625
1024x1024	128	0.0912	0.0091	10.02	0.00001937	0.00390625
512x512	128	0.0977	0.0032	30.53	0.00001580	0.00195312

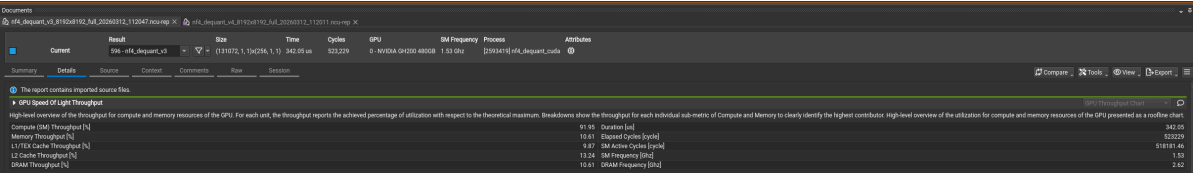
nf4_dequant_v2

半精比float计算快，将中间的变量转化为半精执行。

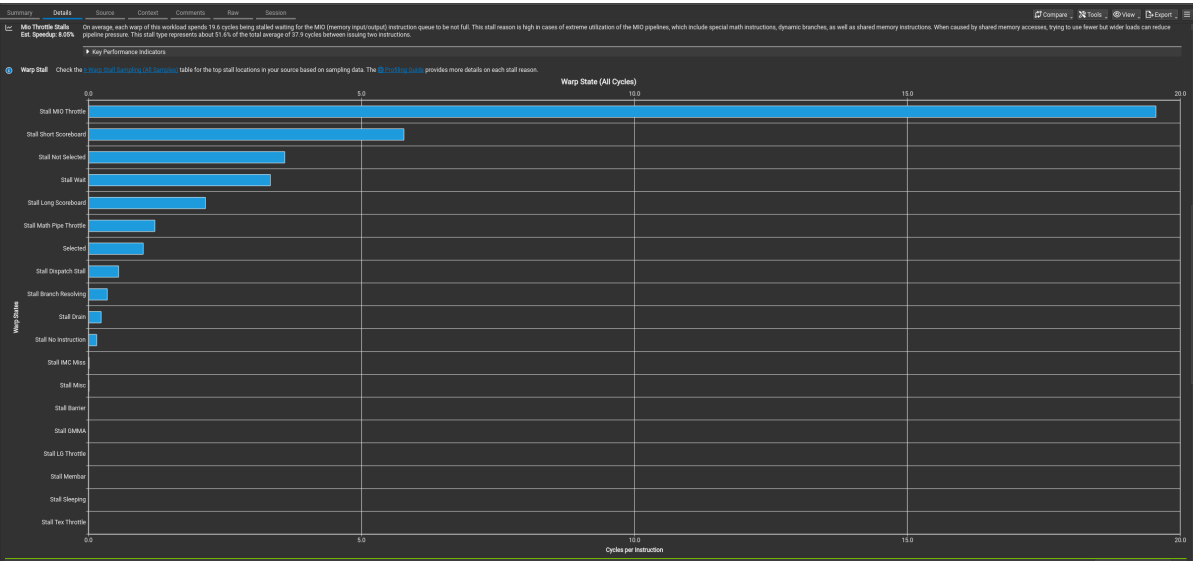
Shape	Block	BnB (ms)	cuda (ms)	Speedup	MAE	Max Diff
4096x4096	128	0.0905	0.0670	1.35	0.00020349	0.00390625
4096x4096	64	0.0891	0.0681	1.31	0.00018179	0.00390625
3421x3146	64	0.0925	0.0437	2.12	0.00017142	0.00390625
7000x7000	128	0.0939	0.1929	0.49	0.00025988	0.00390625
2048x2048	64	0.0910	0.0185	4.92	0.00022352	0.00390625
8192x8192	128	0.0912	0.2631	0.35	0.00022542	0.00390625
6578x1236	64	0.0895	0.0335	2.67	0.00022066	0.00390625
2048x2048	128	0.0925	0.0181	5.11	0.00029397	0.00390625
7000x7000	64	0.0909	0.1984	0.46	0.00024140	0.00390625
256x256	128	0.1064	0.0027	39.41	0.00024271	0.00390625
256x256	64	0.1144	0.0027	42.37	0.00022900	0.00390625
6578x1236	128	0.0876	0.0328	2.67	0.00026894	0.00390625
512x512	64	0.1010	0.0033	30.61	0.00018466	0.00390625
16384x16384	128	0.2973	1.0421	0.29	0.00022352	0.00390625
16384x16384	64	0.3074	1.0718	0.29	0.00019693	0.00390625
8192x8192	64	0.0918	0.2703	0.34	0.00017440	0.00390625
1024x1024	64	0.0960	0.0066	14.55	0.00028801	0.00390625
3421x3146	128	0.0904	0.0426	2.12	0.00026965	0.00390625
1024x1024	128	0.0912	0.0065	14.03	0.00024605	0.00390625
512x512	128	0.0977	0.0032	30.53	0.00029063	0.00390625

nf4_dequant_v3

NF4_LUT（NF4量化的查找表）每次都要从float转化为_half，在核函数启动前提前转化。
下面对8192x8192大小的反量化进行ncu分析，下面各个核函数都是对8192×8192的分析。
计算吞吐率达到90%，而内存吞吐只有10%左右。



查看warp状态，Stall MIO Throttle占主要部分



查找具体是下面两个地方对于内存的读取导致的

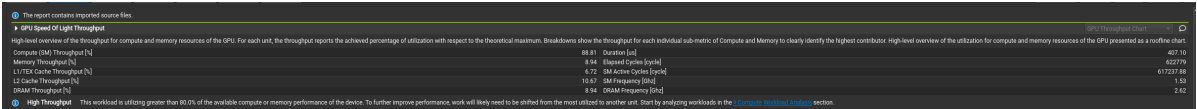


Shape	Block	BnB (ms)	cuda (ms)	Speedup	MAE	Max Diff
4096x4096	128	0.0905	0.0669	1.35	0.00020349	0.00390625
4096x4096	64	0.0891	0.0679	1.31	0.00018179	0.00390625
3421x3146	64	0.0925	0.0438	2.11	0.00017142	0.00390625
7000x7000	128	0.0939	0.1932	0.49	0.00025988	0.00390625
2048x2048	64	0.0910	0.0186	4.89	0.00022352	0.00390625
8192x8192	128	0.0912	0.2636	0.35	0.00022542	0.00390625
6578x1236	64	0.0895	0.0337	2.66	0.00022066	0.00390625
2048x2048	128	0.0925	0.0180	5.14	0.00029397	0.00390625
7000x7000	64	0.0909	0.1987	0.46	0.00024140	0.00390625
256x256	128	0.1064	0.0027	39.41	0.00024271	0.00390625

Shape	Block	BnB (ms)	cuda (ms)	Speedup	MAE	Max Diff
256x256	64	0.1144	0.0028	40.86	0.00022900	0.00390625
6578x1236	128	0.0876	0.0326	2.69	0.00026894	0.00390625
512x512	64	0.1010	0.0035	28.86	0.00018466	0.00390625
16384x16384	128	0.2973	1.0441	0.28	0.00022352	0.00390625
16384x16384	64	0.3074	1.0733	0.29	0.00019693	0.00390625
8192x8192	64	0.0918	0.2710	0.34	0.00017440	0.00390625
1024x1024	64	0.0960	0.0066	14.55	0.00028801	0.00390625
3421x3146	128	0.0904	0.0428	2.11	0.00026965	0.00390625
1024x1024	128	0.0912	0.0065	14.03	0.00024605	0.00390625
512x512	128	0.0977	0.0034	28.74	0.00029063	0.00390625

nf4_dequant_v4

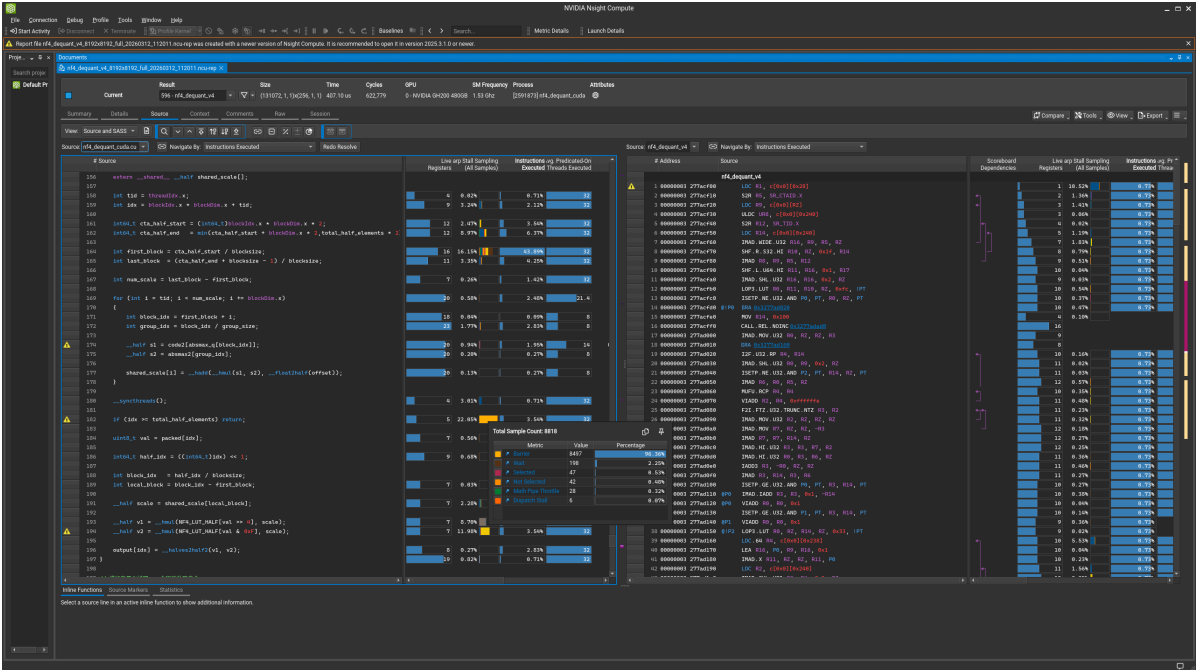
利用共享内存存储scale，但当几个线程在进行存储时其他的线程在等待，时间并没有缩短。
相比于v3，内存吞吐率更低了。



查看共享内存状态大量bank conflicts

	Instructions	Requests	Wavefronts	% Peak	Bank Conflicts
Shared Load	1048576	1048576	1060174	1.29	11598
Shared Load Matrix	0	0			
Shared Store	131072	131072	175759	0.21	44687
Shared Store Matrix	0	0			
Shared Store From Global Load	0	0	0	0	0
Shared Atomic	0	0	0	0	0
Other	-	-	1169220	1.42	0
Total	1179648	1179648	2405153	2.92	56285

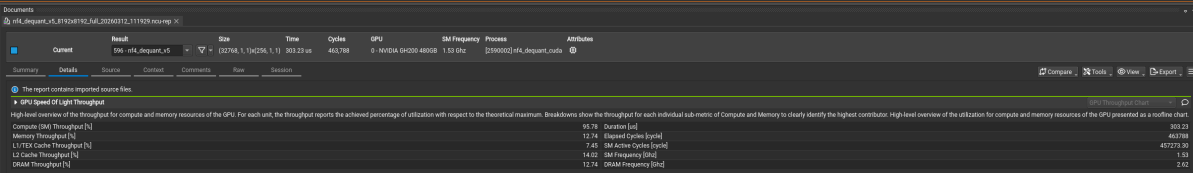
查看使用共享内存存在大量barrier



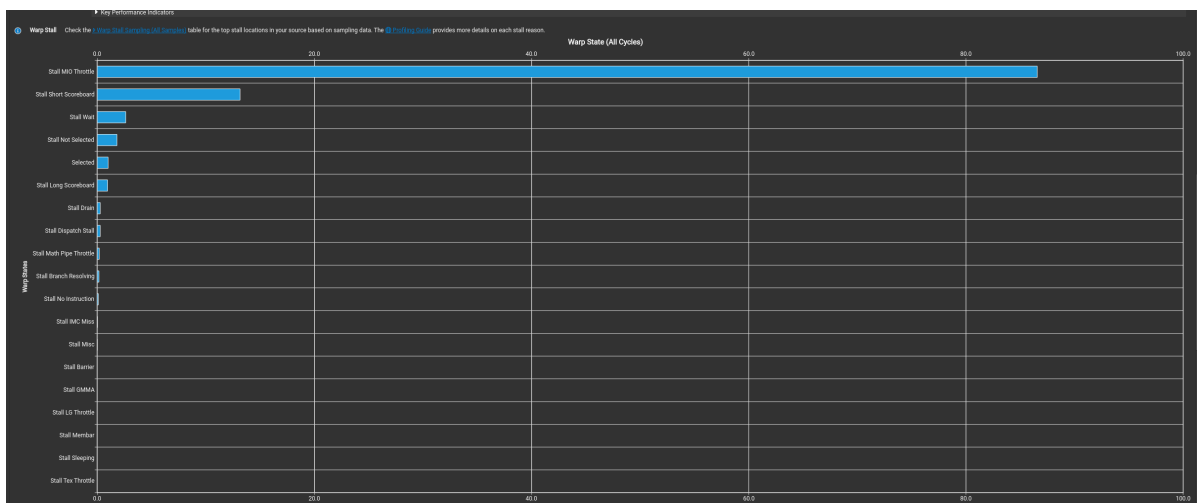
Shape	Block	BnB (ms)	cuda (ms)	Speedup	MAE	Max Diff
4096x4096	128	0.0905	0.0802	1.13	0.00020349	0.00390625
4096x4096	64	0.0891	0.0821	1.09	0.00018179	0.00390625
3421x3146	64	0.0925	0.0531	1.74	0.00017142	0.00390625
7000x7000	128	0.0939	0.2308	0.41	0.00025988	0.00390625
2048x2048	64	0.0910	0.0224	4.06	0.00022352	0.00390625
8192x8192	128	0.0912	0.3140	0.29	0.00022542	0.00390625
6578x1236	64	0.0895	0.0408	2.19	0.00022066	0.00390625
2048x2048	128	0.0925	0.0219	4.22	0.00029397	0.00390625
7000x7000	64	0.0909	0.2386	0.38	0.00024140	0.00390625
256x256	128	0.1064	0.0033	32.24	0.00024271	0.00390625
256x256	64	0.1144	0.0032	35.75	0.00022900	0.00390625
6578x1236	128	0.0876	0.0397	2.21	0.00026894	0.00390625
512x512	64	0.1010	0.0040	25.25	0.00018466	0.00390625
16384x16384	128	0.2973	1.2404	0.24	0.00022352	0.00390625
16384x16384	64	0.3074	1.2822	0.24	0.00019693	0.00390625
8192x8192	64	0.0918	0.3244	0.28	0.00017440	0.00390625
1024x1024	64	0.0960	0.0080	12.00	0.00028801	0.00390625
3421x3146	128	0.0904	0.0518	1.75	0.00026965	0.00390625
1024x1024	128	0.0912	0.0079	11.54	0.00024605	0.00390625
512x512	128	0.0977	0.0040	24.42	0.00029063	0.00390625

nf4_dequant_v5

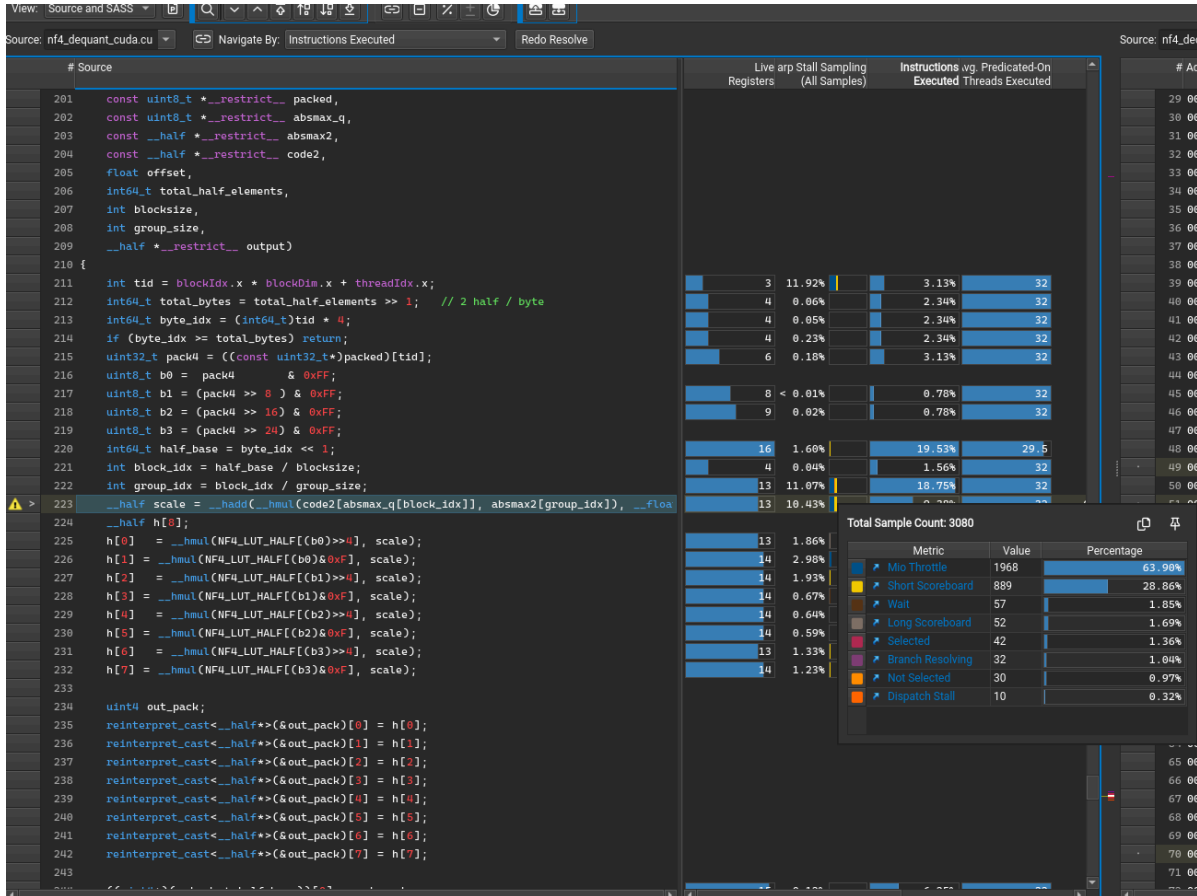
进行向量化读取，一个程序处理多个元素
现在的计算吞吐达到95%，但内存吞吐还是很低。

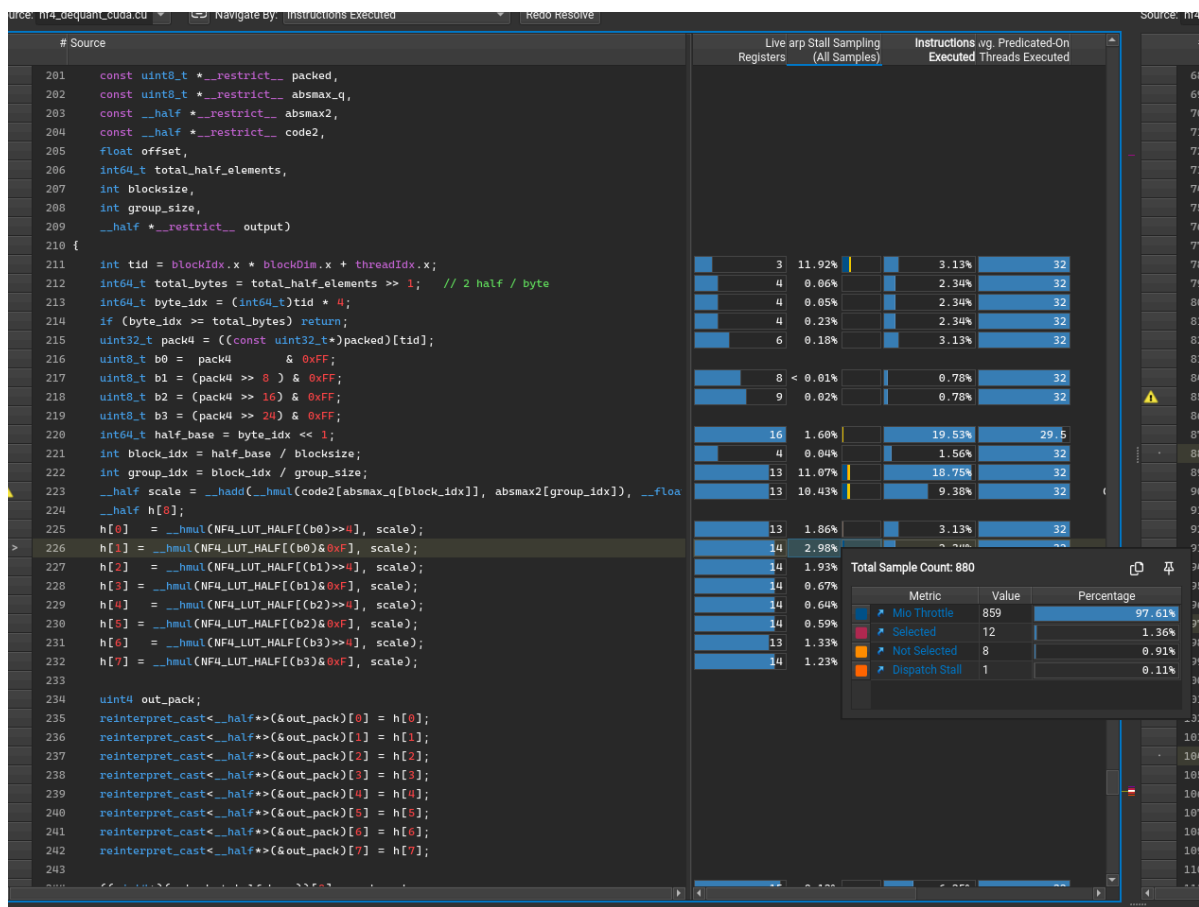


还是大量存在Stall MIO Throttle



对于NF4_LUT还是有大量的Stall MIO Throtle，还有对于计算scale也有Stall MIO Throtle



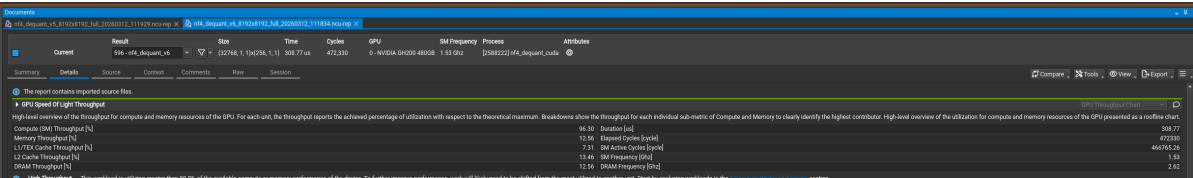


Shape	Block	BnB (ms)	cuda (ms)	Speedup	MAE	Max Diff
4096x4096	128	0.0905	0.0847	1.07	0.00020349	0.00390625
4096x4096	64	0.0891	0.0928	0.96	0.00018179	0.00390625
3421x3146	64	0.0925	0.0452	2.05	0.00017142	0.00390625
7000x7000	128	0.0939	0.3667	0.26	0.00025988	0.00390625
2048x2048	64	0.0910	0.0197	4.62	0.00022352	0.00390625
8192x8192	128	0.0912	0.2261	0.40	0.00022542	0.00390625
6578x1236	64	0.0895	0.0401	2.23	0.00022066	0.00390625
2048x2048	128	0.0925	0.0193	4.79	0.00029397	0.00390625
7000x7000	64	0.0909	0.1721	0.53	0.00024140	0.00390625
256x256	128	0.1064	0.0060	17.73	0.00024271	0.00390625
256x256	64	0.1144	0.0062	18.45	0.00022900	0.00390625
6578x1236	128	0.0876	0.0433	2.02	0.00026894	0.00390625
512x512	64	0.1010	0.0034	29.71	0.00018466	0.00390625
16384x16384	128	0.2973	1.4397	0.21	0.00022352	0.00390625
16384x16384	64	0.3074	1.0827	0.28	0.00019693	0.00390625
8192x8192	64	0.0918	0.2343	0.39	0.00017440	0.00390625

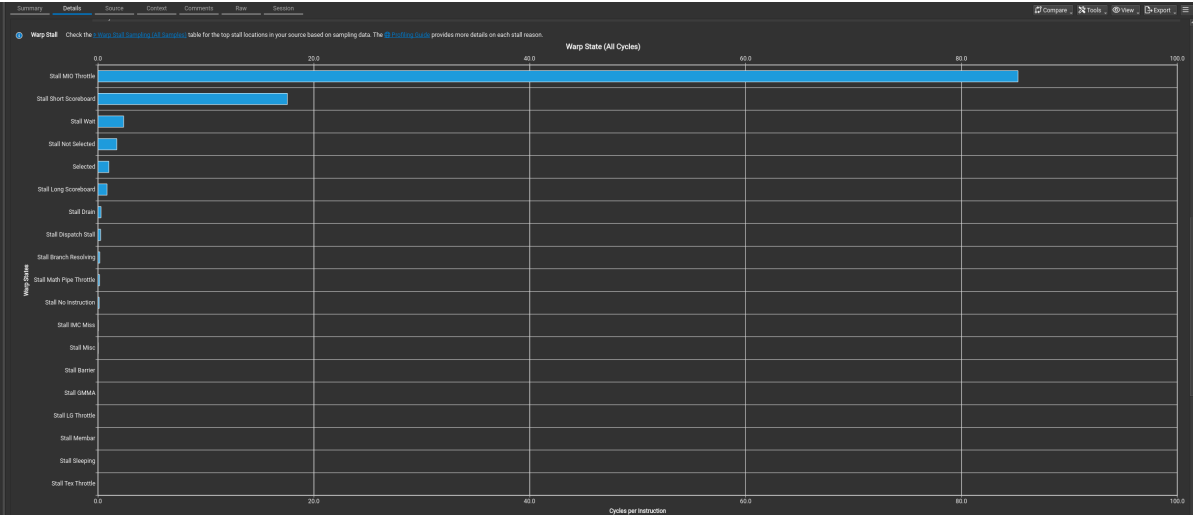
Shape	Block	BnB (ms)	cuda (ms)	Speedup	MAE	Max Diff
1024x1024	64	0.0960	0.0094	10.21	0.00028801	0.00390625
3421x3146	128	0.0904	0.0445	2.03	0.00026965	0.00390625
1024x1024	128	0.0912	0.0058	15.72	0.00024605	0.00390625
512x512	128	0.0977	0.0067	14.58	0.00029063	0.00390625

nf4_dequant_v6:

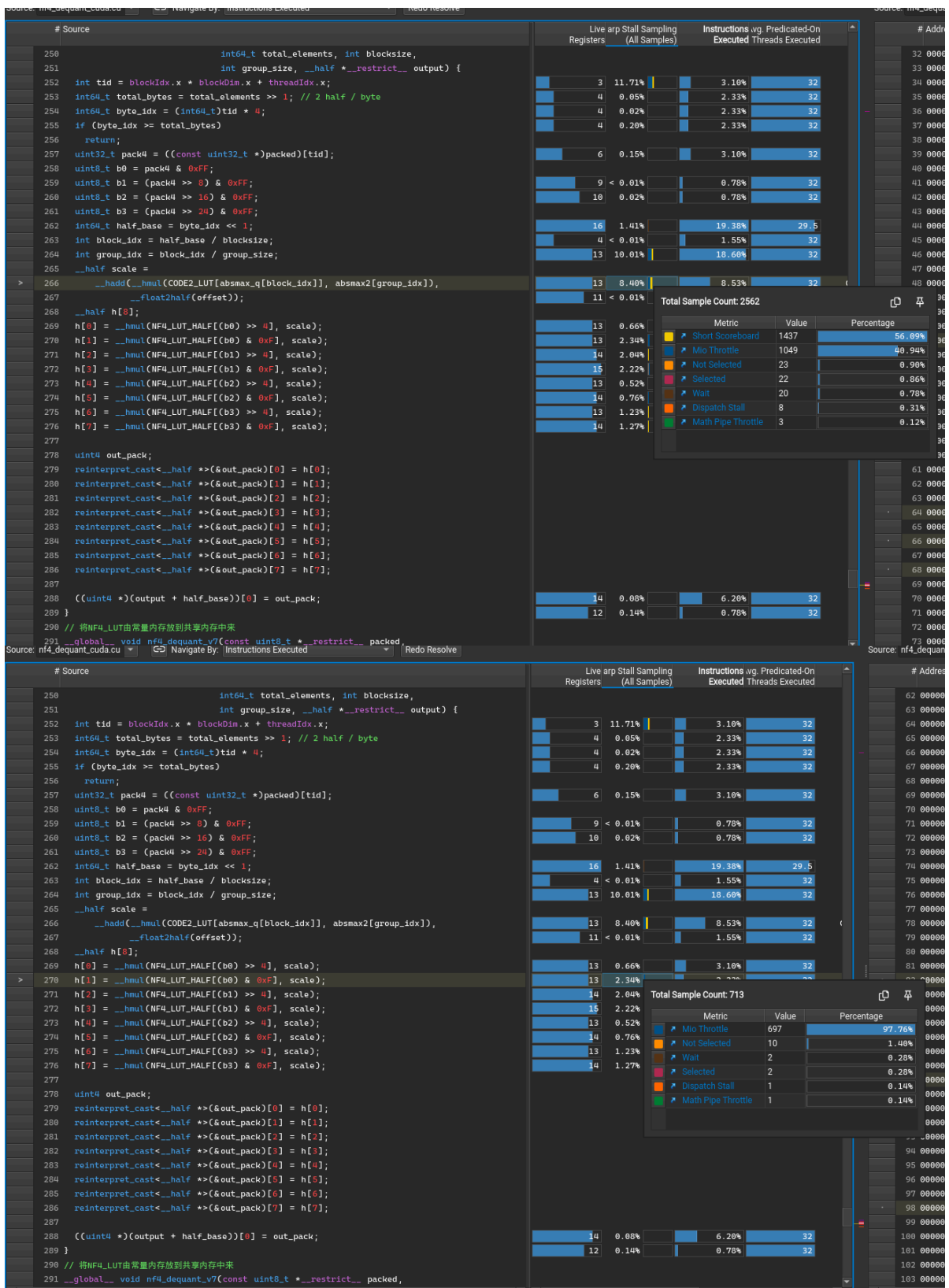
将code也放入常量内存，减少从全局内存进行读取
现在的计算吞吐达到96%，但内存吞吐还是很低。



还是大量存在Stall MIO Throttle



对于NF4_LUT还是有大量的Stall MIO Throttle，还有对于code2的Stall MIO Throttle有所减少。



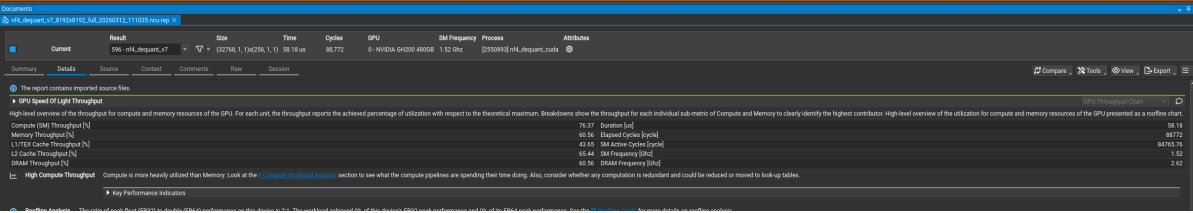
Shape	Block	BnB (ms)	cuda (ms)	Speedup	MAE	Max Diff
4096x4096	128	0.0905	0.0594	1.52	0.00020349	0.00390625
4096x4096	64	0.0891	0.0621	1.43	0.00018179	0.00390625
3421x3146	64	0.0925	0.0404	2.29	0.00017142	0.00390625
7000x7000	128	0.0939	0.1671	0.56	0.00025988	0.00390625
2048x2048	64	0.0910	0.0176	5.17	0.00022352	0.00390625

Shape	Block	BnB (ms)	cuda (ms)	Speedup	MAE	Max Diff
8192x8192	128	0.0912	0.2273	0.40	0.00022542	0.00390625
6578x1236	64	0.0895	0.0316	2.83	0.00022066	0.00390625
2048x2048	128	0.0925	0.0168	5.51	0.00029397	0.00390625
7000x7000	64	0.0909	0.1754	0.52	0.00024140	0.00390625
256x256	128	0.1064	0.0035	30.40	0.00024271	0.00390625
256x256	64	0.1144	0.0035	32.69	0.00022900	0.00390625
6578x1236	128	0.0876	0.0300	2.92	0.00026894	0.00390625
512x512	64	0.1010	0.0035	28.86	0.00018466	0.00390625
16384x16384	128	0.2973	0.8980	0.33	0.00022352	0.00390625
16384x16384	64	0.3074	0.9441	0.33	0.00019693	0.00390625
8192x8192	64	0.0918	0.2389	0.38	0.00017440	0.00390625
1024x1024	64	0.0960	0.0062	15.48	0.00028801	0.00390625
3421x3146	128	0.0904	0.0385	2.35	0.00026965	0.00390625
1024x1024	128	0.0912	0.0060	15.20	0.00024605	0.00390625
512x512	128	0.0977	0.0036	27.14	0.00029063	0.00390625

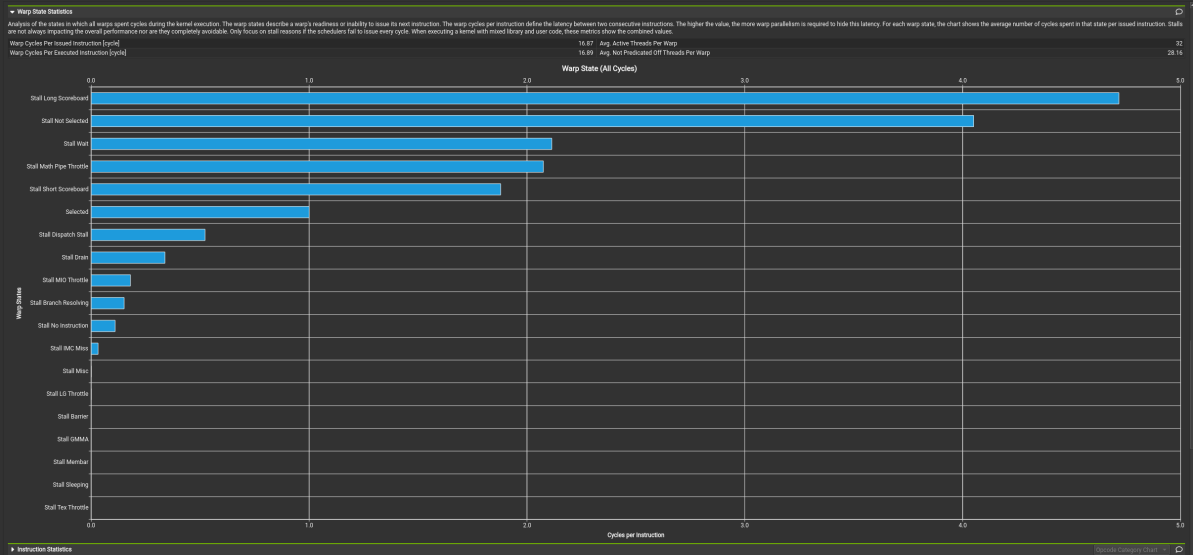
nf4_dequant_v7:

将NF4_LUT由常量内存放到共享内存中来，速度比v6版本大大提高了

内存吞吐达到60%，且计算时间大大减少



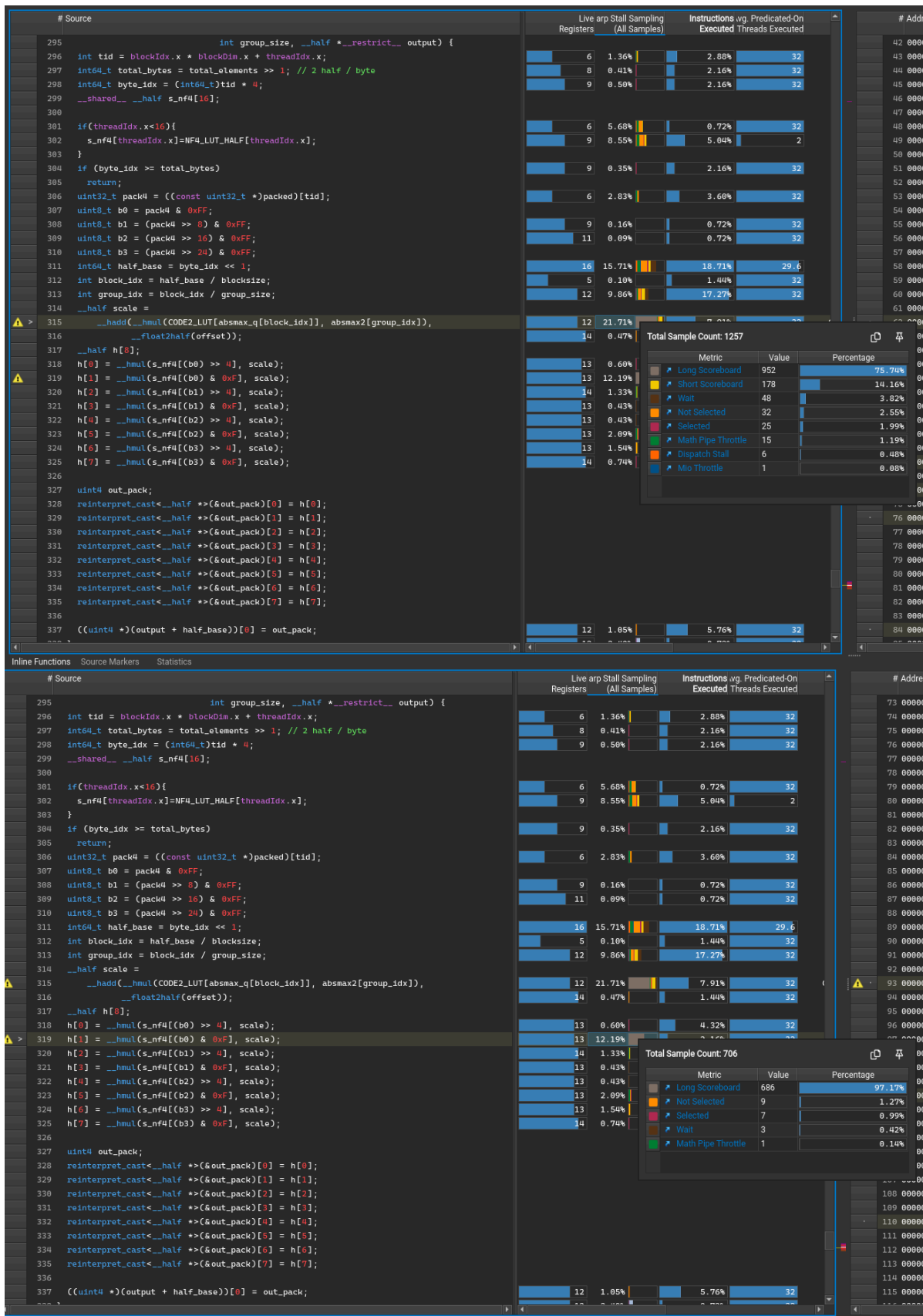
Stall MIO Throttle大大减少



引入共享内存后出现bank conflicts

	Instructions	Requests	Wavefronts	% Peak	Bank Conflicts
Shared Load	2097152	2097152	2246862	19.23	149710
Shared Load Matrix	0	0			
Shared Store	32768	32768	38530	0.33	5762
Shared Store Matrix	0	0			
Shared Store From Global Load	0	0	0	0	0
Shared Atomic	0	0	0	0	0
Other	-	-	490079	4.20	0
Total	2129920	2129920	2775471	23.76	155472

对于NF4_LUT已经计算scale的Stall MIO Throttle消失



Shape	Block	BnB (ms)	cuda (ms)	Speedup	MAE	Max Diff
4096x4096	128	0.0905	0.0131	6.91	0.00020349	0.00390625
4096x4096	64	0.0891	0.0134	6.65	0.00018179	0.00390625
3421x3146	64	0.0925	0.0087	10.63	0.00017142	0.00390625

Shape	Block	BnB (ms)	cuda (ms)	Speedup	MAE	Max Diff
7000x7000	128	0.0939	0.0449	2.09	0.00025988	0.00390625
2048x2048	64	0.0910	0.0047	19.36	0.00022352	0.00390625
8192x8192	128	0.0912	0.0581	1.57	0.00022542	0.00390625
6578x1236	64	0.0895	0.0073	12.26	0.00022066	0.00390625
2048x2048	128	0.0925	0.0049	18.88	0.00029397	0.00390625
7000x7000	64	0.0909	0.0418	2.17	0.00024140	0.00390625
256x256	128	0.1064	0.0028	38.00	0.00024271	0.00390625
256x256	64	0.1144	0.0028	40.86	0.00022900	0.00390625
6578x1236	128	0.0876	0.0070	12.51	0.00026894	0.00390625
512x512	64	0.1010	0.0028	36.07	0.00018466	0.00390625
16384x16384	128	0.2973	0.2156	1.38	0.00022352	0.00390625
16384x16384	64	0.3074	0.2292	1.34	0.00019693	0.00390625
8192x8192	64	0.0918	0.0593	1.55	0.00017440	0.00390625
1024x1024	64	0.0960	0.0031	30.97	0.00028801	0.00390625
3421x3146	128	0.0904	0.0087	10.39	0.00026965	0.00390625
1024x1024	128	0.0912	0.0029	31.45	0.00024605	0.00390625
512x512	128	0.0977	0.0028	34.89	0.00029063	0.00390625

英伟达平台适配结果

对于GH200对nf4_dequant_v7进行支持。执行结果权重文件为cuda_results目录下的.fp16为后缀的文件，log文件记录带宽及其执行时间。加速比在nv/comparison_cuda_results.md

Shape	Block	BnB (ms)	cuda (ms)	Speedup	MAE	Max Diff
4096x4096	128	0.0905	0.0131	6.91	0.00020349	0.00390625
4096x4096	64	0.0891	0.0134	6.65	0.00018179	0.00390625
3421x3146	64	0.0925	0.0087	10.63	0.00017142	0.00390625
7000x7000	128	0.0939	0.0449	2.09	0.00025988	0.00390625
2048x2048	64	0.0910	0.0047	19.36	0.00022352	0.00390625
8192x8192	128	0.0912	0.0581	1.57	0.00022542	0.00390625
6578x1236	64	0.0895	0.0073	12.26	0.00022066	0.00390625

Shape	Block	BnB (ms)	cuda (ms)	Speedup	MAE	Max Diff
2048x2048	128	0.0925	0.0049	18.88	0.00029397	0.00390625
7000x7000	64	0.0909	0.0418	2.17	0.00024140	0.00390625
256x256	128	0.1064	0.0028	38.00	0.00024271	0.00390625
256x256	64	0.1144	0.0028	40.86	0.00022900	0.00390625
6578x1236	128	0.0876	0.0070	12.51	0.00026894	0.00390625
512x512	64	0.1010	0.0028	36.07	0.00018466	0.00390625
16384x16384	128	0.2973	0.2156	1.38	0.00022352	0.00390625
16384x16384	64	0.3074	0.2292	1.34	0.00019693	0.00390625
8192x8192	64	0.0918	0.0593	1.55	0.00017440	0.00390625
1024x1024	64	0.0960	0.0031	30.97	0.00028801	0.00390625
3421x3146	128	0.0904	0.0087	10.39	0.00026965	0.00390625
1024x1024	128	0.0912	0.0029	31.45	0.00024605	0.00390625
512x512	128	0.0977	0.0028	34.89	0.00029063	0.00390625

沐曦平台程序执行及性能分析

```
cd mx
./run_all.sh 对于程序进行编译和对于所有权重文件运行
python compare_results.py 进行性能和误差分析
```

沐曦适配结果

对于沐曦S5000平台对nf4_dequant_v7进行支持，在大矩阵的表现较差。执行结果权重文件为mx_results目录下的.fp16为后缀的文件，log文件记录带宽及其执行时间。加速比在mx/comparison_mx_results.md

Shape	Block	BnB (ms)	MX (ms)	Speedup	MAE	Max Diff
1024x1024	128	0.0912	0.0248	3.68	0.00024605	0.00390625
1024x1024	64	0.0960	0.0169	5.68	0.00028801	0.00390625
16384x16384	128	0.2973	0.9282	0.32	0.00022352	0.00390625
16384x16384	64	0.3074	1.0106	0.30	0.00019693	0.00390625
2048x2048	128	0.0925	0.0325	2.85	0.00029397	0.00390625
2048x2048	64	0.0910	0.0281	3.24	0.00022352	0.00390625

Shape	Block	BnB (ms)	MX (ms)	Speedup	MAE	Max Diff
256x256	128	0.1064	0.0093	11.44	0.00024271	0.00390625
256x256	64	0.1144	0.0094	12.17	0.00022900	0.00390625
3421x3146	128	0.0904	0.0509	1.78	0.00026965	0.00390625
3421x3146	64	0.0925	0.0517	1.79	0.00017142	0.00390625
4096x4096	128	0.0905	0.0813	1.11	0.00020349	0.00390625
4096x4096	64	0.0891	0.0735	1.21	0.00018179	0.00390625
512x512	128	0.0977	0.0154	6.34	0.00029063	0.00390625
512x512	64	0.1010	0.0154	6.56	0.00018466	0.00390625
6578x1236	128	0.0876	0.0402	2.18	0.00026894	0.00390625
6578x1236	64	0.0895	0.0485	1.85	0.00022066	0.00390625
7000x7000	128	0.0939	0.1881	0.50	0.00025988	0.00390625
7000x7000	64	0.0909	0.1905	0.48	0.00024140	0.00390625
8192x8192	128	0.0912	0.2358	0.39	0.00022542	0.00390625
8192x8192	64	0.0918	0.2625	0.35	0.00017440	0.00390625

摩尔线程程序执行及性能分析

```
cd mexc
// 进行编译
$MUSA_ROOT/bin/mcc -O3 -std=c++17 \
-I$MUSA_ROOT/include \
-L$MUSA_ROOT/lib \
-L/usr/lib/gcc/x86_64-linux-gnu/11 \
-lmusart -lstdc++ \
nf4_dequant_musa.mu -o nf4_musa
// 提交任务执行权重文件
sbatch test.sbatch
// 进行性能和误差分析
python compare_results.py
```

摩尔线程适配结果

对于S5000对nf4_dequant_v7进行支持，在中大矩阵的表现差，且有效带宽在最大时也只有229GB/s。执行结果权重文件为mexc_results目录下的.fp16为后缀的文件，log文件记录带宽及其执行时间。加速比在mexc/comparison_musa_results.md

Shape	Block	BnB (ms)	MUSA (ms)	Speedup	MAE	MSE	Max Diff
2048x2048	128	0.0925	0.0435	2.13	0.00029397	0.00000030	0.00390625

Shape	Block	BnB (ms)	MUSA (ms)	Speedup	MAE	MSE	Max Diff
512x512	64	0.1010	0.0080	12.62	0.00018466	0.00000018	0.00390625
256x256	64	0.1144	0.0074	15.46	0.00022900	0.00000024	0.00390625
3421x3146	64	0.0925	0.1168	0.79	0.00017142	0.00000012	0.00390625
7000x7000	64	0.0909	0.5581	0.16	0.00024140	0.00000024	0.00390625
7000x7000	128	0.0939	0.5360	0.18	0.00025988	0.00000024	0.00390625
2048x2048	64	0.0910	0.0456	2.00	0.00022352	0.00000018	0.00390625
6578x1236	128	0.0876	0.0832	1.05	0.00026894	0.00000024	0.00390625
1024x1024	128	0.0912	0.0137	6.66	0.00024605	0.00000024	0.00390625
256x256	128	0.1064	0.0077	13.82	0.00024271	0.00000024	0.00390625
8192x8192	64	0.0918	0.7726	0.12	0.00017440	0.00000012	0.00390625
4096x4096	64	0.0891	0.1832	0.49	0.00018179	0.00000018	0.00390625
8192x8192	128	0.0912	0.7425	0.12	0.00022542	0.00000018	0.00390625
1024x1024	64	0.0960	0.0145	6.62	0.00028801	0.00000030	0.00390625
6578x1236	64	0.0895	0.0867	1.03	0.00022066	0.00000018	0.00390625
16384x16384	64	0.3074	3.3083	0.09	0.00019693	0.00000018	0.00390625
512x512	128	0.0977	0.0078	12.53	0.00029063	0.00000030	0.00390625
4096x4096	128	0.0905	0.1756	0.52	0.00020349	0.00000018	0.00390625
3421x3146	128	0.0904	0.1103	0.82	0.00026965	0.00000024	0.00390625
16384x16384	128	0.2973	3.1883	0.09	0.00022352	0.00000018	0.00390625

未来可继续提升的地方

解决v7版本存在的bank conflicts,v7版本在国产GPU上执行较差，找到原因解决。